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COMMENT GUIDELINES

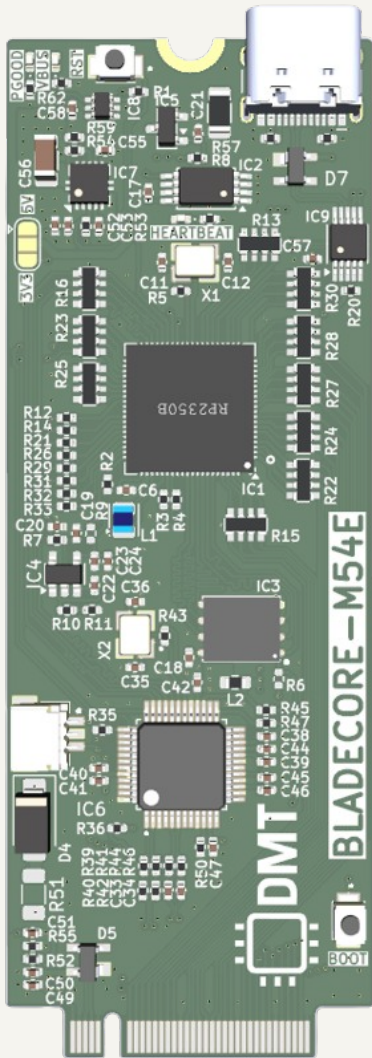
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

NOTES

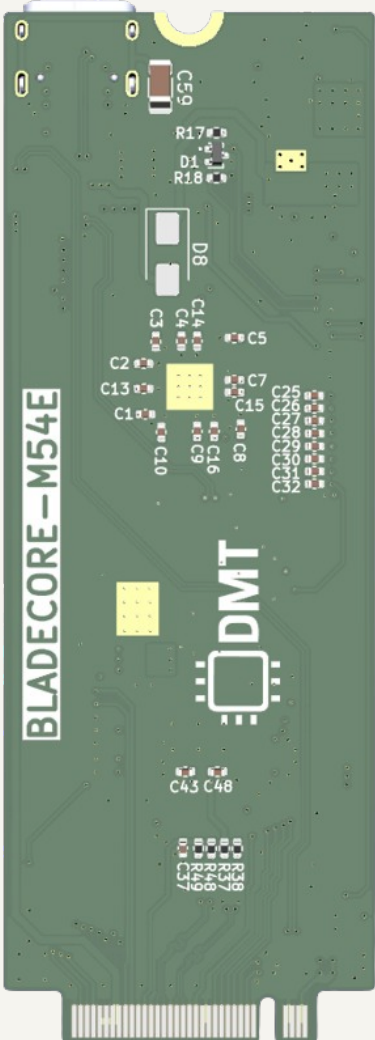
Not fitted components are marked as ~~X~~

PCB PREVIEW

TOP VIEW



BOTTOM VIEW



 Date: 2026-01-31 Sheet Title: BladeCore-M54E	Board Name: BladeCore		Project Name: M54	
	File Name: BladeCore-M54E.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 9

The diagram illustrates the internal architecture of the 2280 SIZE M.2 M-KEY CARD. The central component is the **RP2354B** microcontroller, which is connected to various external interfaces and components.

Power and Voltage Regulation:

- A **+5V** input is connected to the **USB INTERFACE** and the **LDO** (Low Dropout Regulator).
- The **LDO** outputs **+3.3V** to the **RP2354B** and the **VBUS/3.3V/5V** line.
- A **3.0V VREF** is provided to the **ADC** (Analog-to-Digital Converter).
- VUSB** is also connected to the **ADC**.

USB and Data Interfaces:

- The **USB INTERFACE** is connected to the **RP2354B** via the **ONBOARD USB** and **CARD EDGE USB** pins.
- The **RP2354B** has a **USB MUX** (Multiplexer) that can route data between the **ONBOARD USB** and **CARD EDGE USB**.
- The **RP2354B** is connected to the **IO** (Input/Output) pins, which are further connected to the **GPIO0-27** pins.

Internal Components and Connections:

- The **RP2354B** is connected to the **ADC** (Analog-to-Digital Converter) via the **ADC AVDD** and **GPIO47** pins.
- The **RP2354B** is connected to the **GPIO46** pin, which is further connected to the **GPIO40-45** pins.
- The **RP2354B** is connected to the **GPIO32-33** pins, which are further connected to the **GPIO36** pin.
- The **RP2354B** is connected to the **GPIO28-31** and **GPIO34-35** pins, which are further connected to the **GPIO36** pin.
- The **RP2354B** is connected to the **GPIO0-27** pins, which are further connected to the **GPIO36** pin.
- The **RP2354B** is connected to the **GPIO36** pin, which is further connected to the **GPIO40-45** pins.
- The **RP2354B** is connected to the **GPIO40-45** pins, which are further connected to the **GPIO46** pin.
- The **RP2354B** is connected to the **GPIO46** pin, which is further connected to the **GPIO47** pin.
- The **RP2354B** is connected to the **GPIO47** pin, which is further connected to the **GPIO46** pin.

External Components and Connections:

- The **RP2354B** is connected to the **W5500 100BASE ETHERNET CONTROLLER** via the **SPI1** and **I2C0** pins.
- The **RP2354B** is connected to the **256KB EEPROM** via the **I2C0** pin.
- The **RP2354B** is connected to the **W25Q128JVPIQ 8MB FLASH** via the **QSPI** pin.
- The **RP2354B** is connected to the **HEARTBEAT LED** via the **IO** pin.
- The **RP2354B** is connected to the **ADC0-ADC5** via the **ADC** pin.
- The **RP2354B** is connected to the **SWD** (Serial Wire Debug) pin, which is further connected to the **SWD HEADER**.

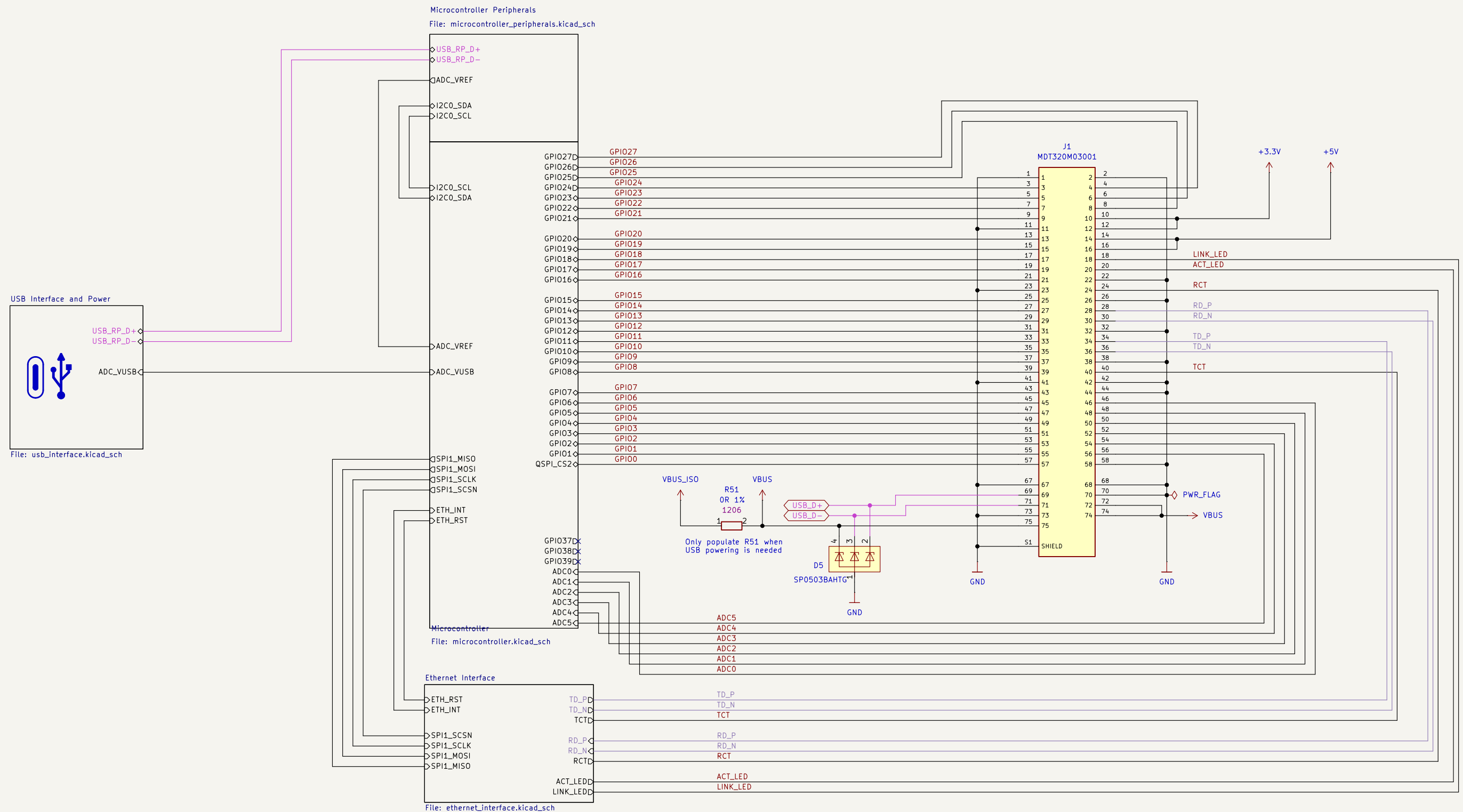
Other Features:

- The **RP2354B** has a **STATUS LEDS** (Status LEDs) connected to it.
- The **RP2354B** has a **TD+/TD- RD+/RD- TCT, RCT ACTLED, LINKLED** (Transmit/Receive Data, Transmit/Receive Control, Active LED, Link LED) connected to it.

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	480 mA
Typical current consumption	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV

DMT	Board Name: BladeCore		Project Name: M54	
	File Name: Block Diagram.kicad_sch		Revision: 1.0.0	Variant:
Sheet Title: Block Diagram	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

[3] Project Architecture



Date: 2026-01-31

Sheet Title:
Project Architecture

Board Name:	BladeCore
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File Name:
Project Architecture.kicad_sch

Company:
DvidMakesThings

Project Name:
M54

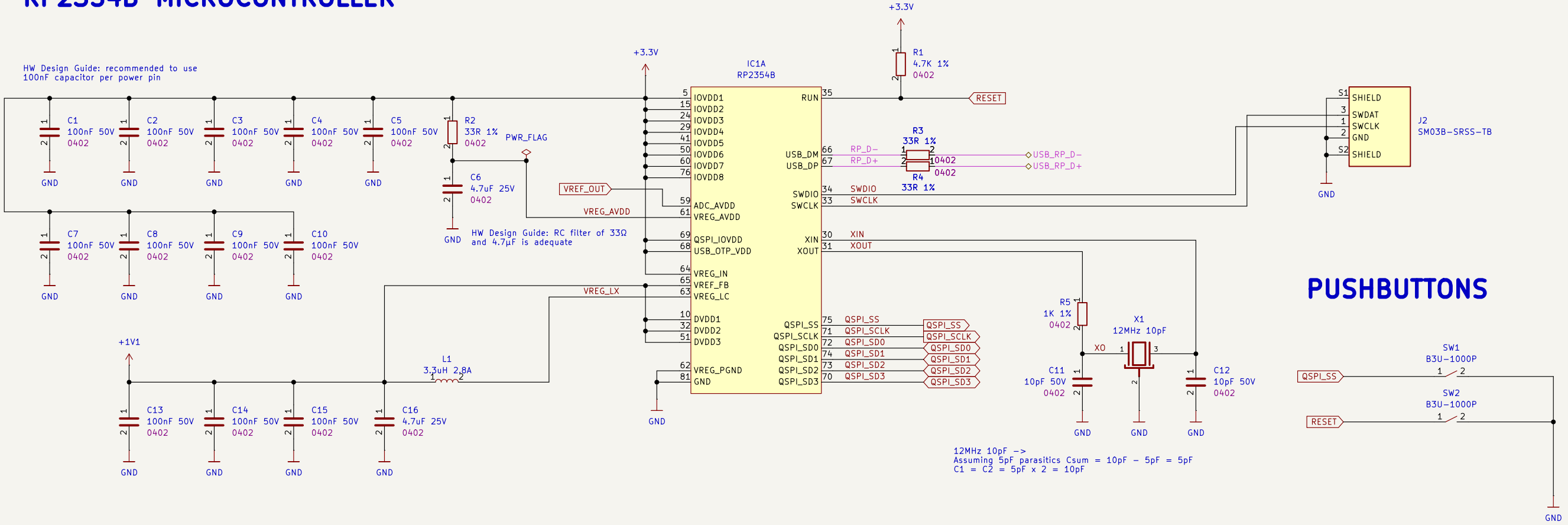
Revision:	Variant:
1.0.0	

Size:	Sheet:
A3	3 of 9

[4] Microcontroller Peripherals

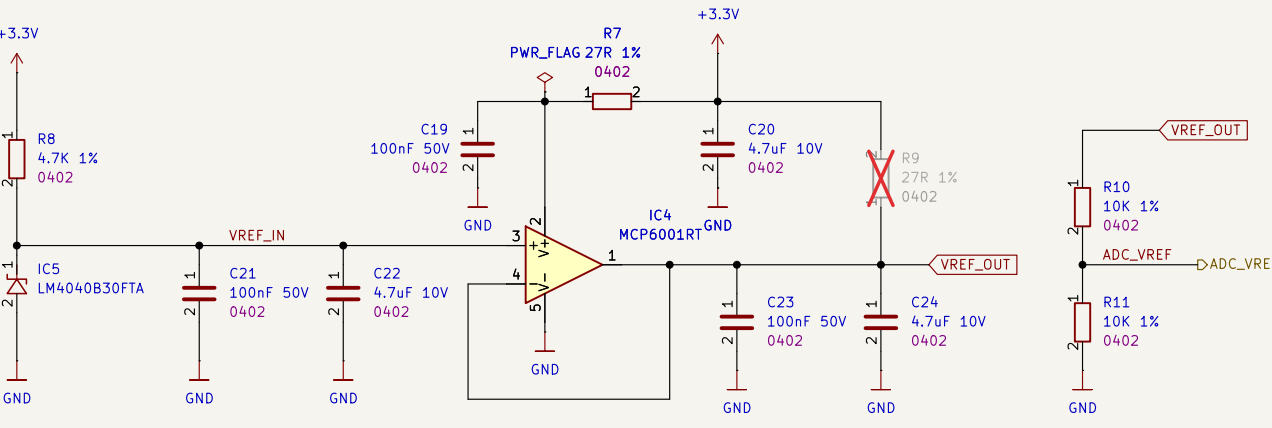
RP2354B MICROCONTROLLER

HW Design Guide: recommended to use 100nF capacitor per power pin

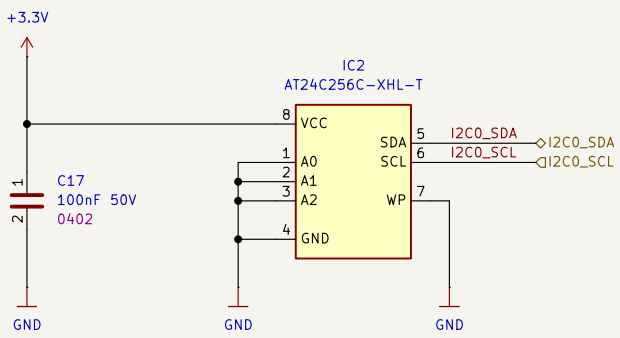


PUSHBUTTONS

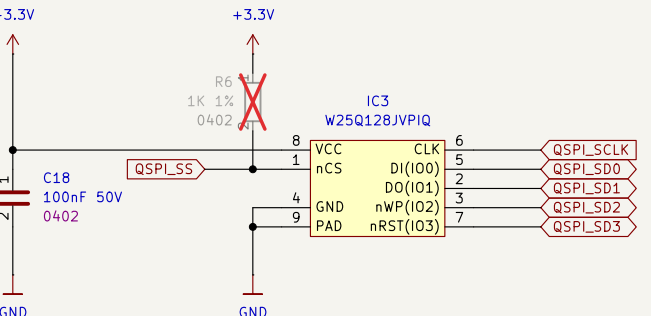
3.0V ADC VOLTAGE REFERENCE



EXTERNAL EEPROM



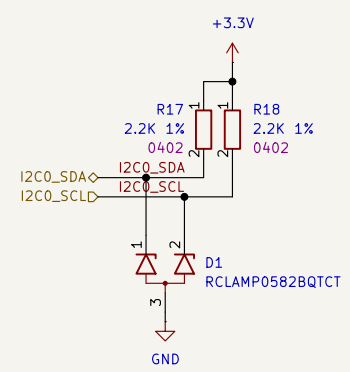
EXTERNAL FLASH



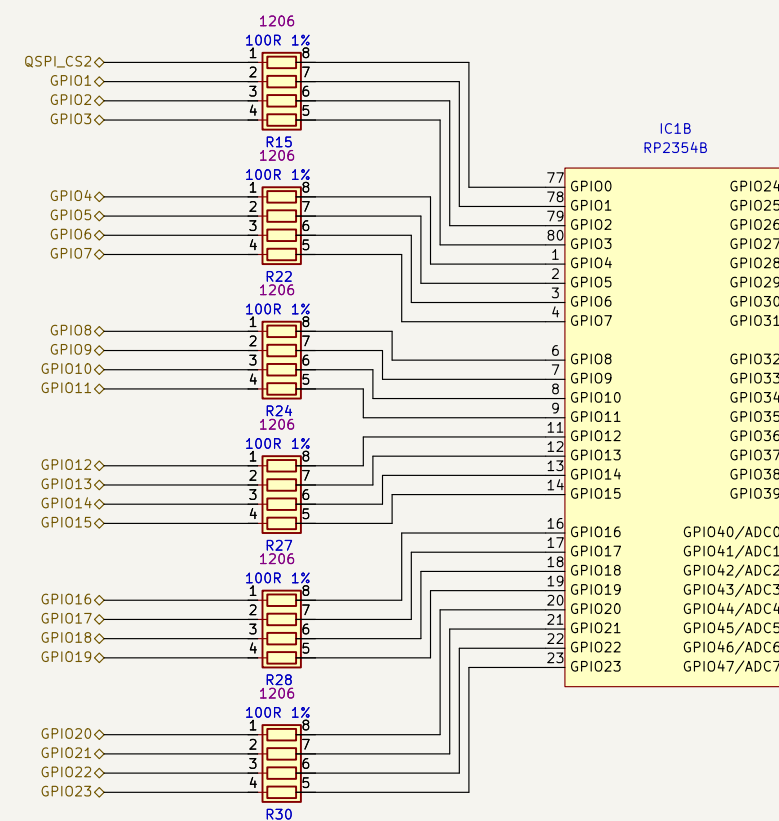
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	File Name: microcontroller_peripherals.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 4 of 9

[5] Microcontroller

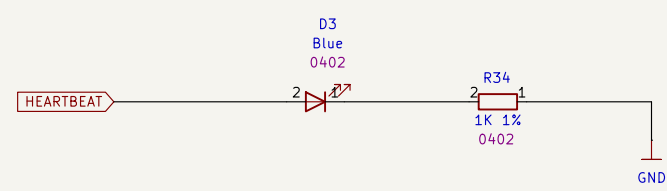
I2C PULLUP



RP2354B GPIO ASSIGNMENT



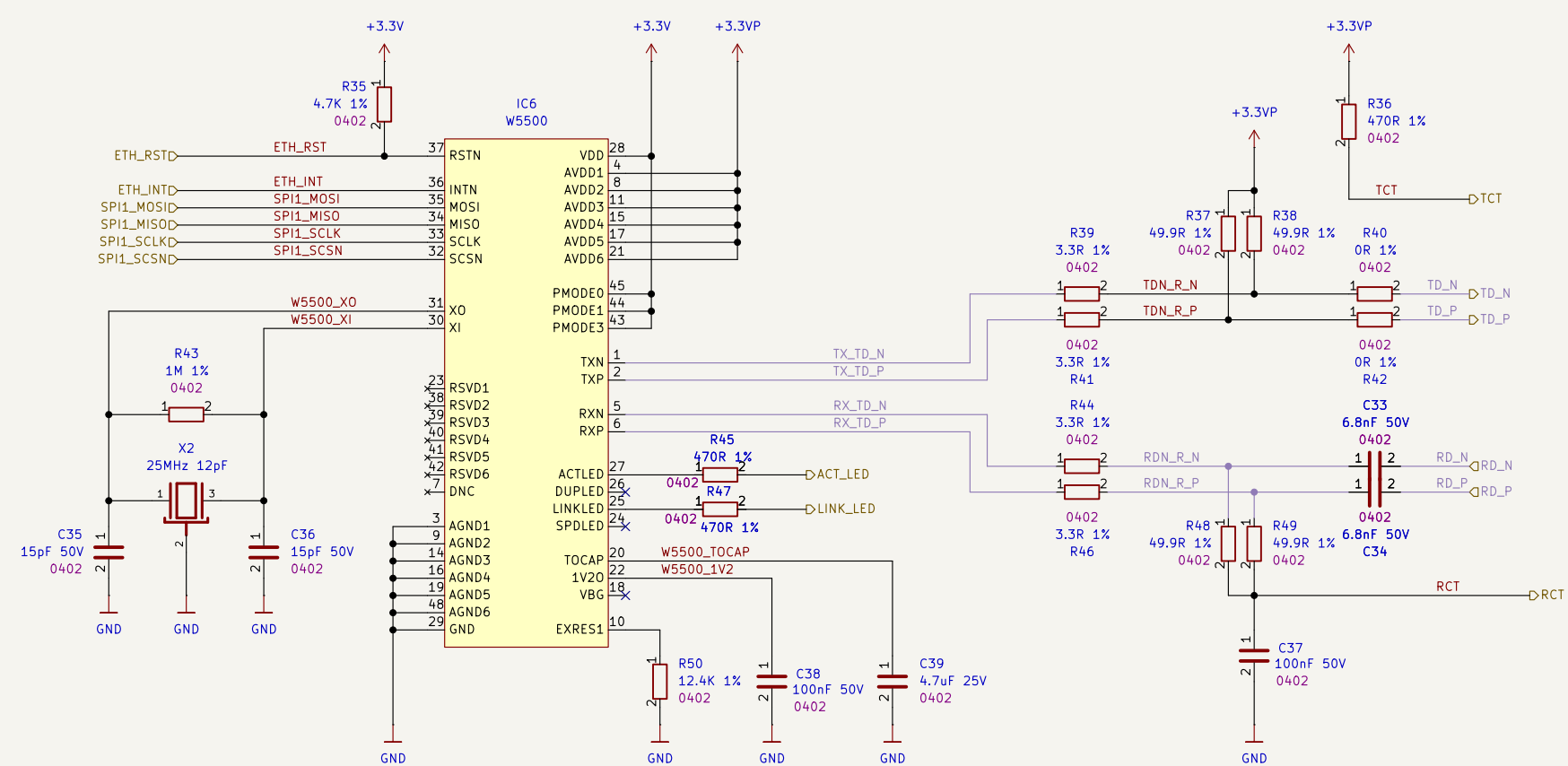
HEARTBEAT LED



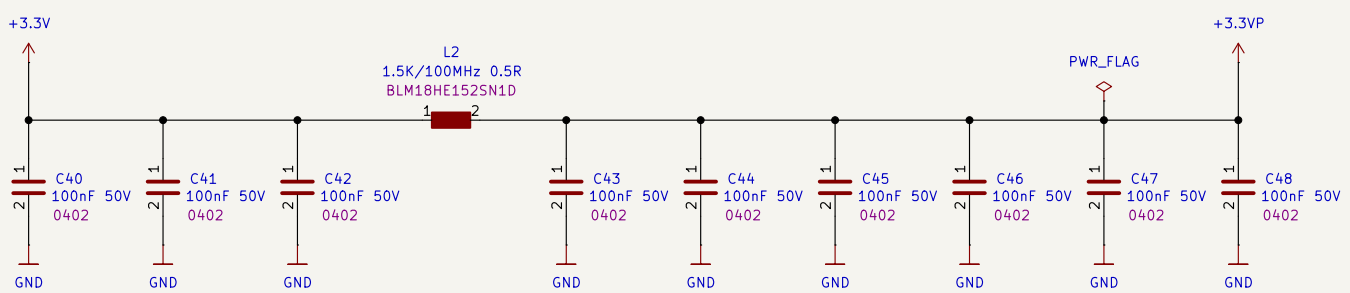
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	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 5 of 9

[6] Ethernet Interface

W5500 ETHERNET CONTROLLER WITH PHY



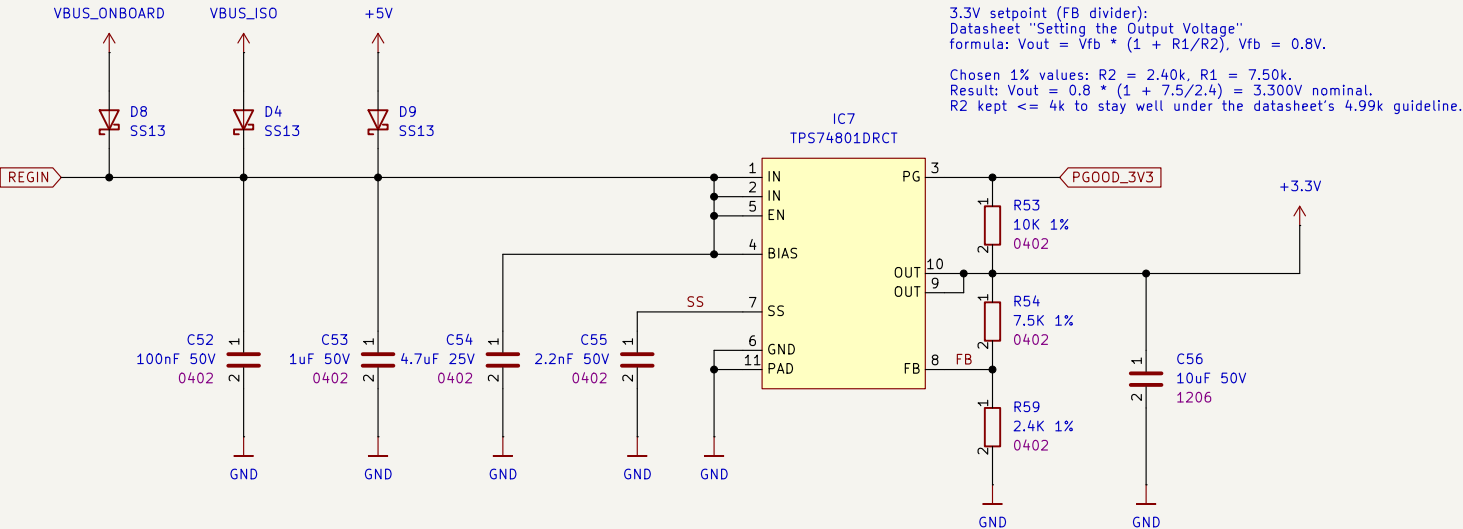
DECOUPLING



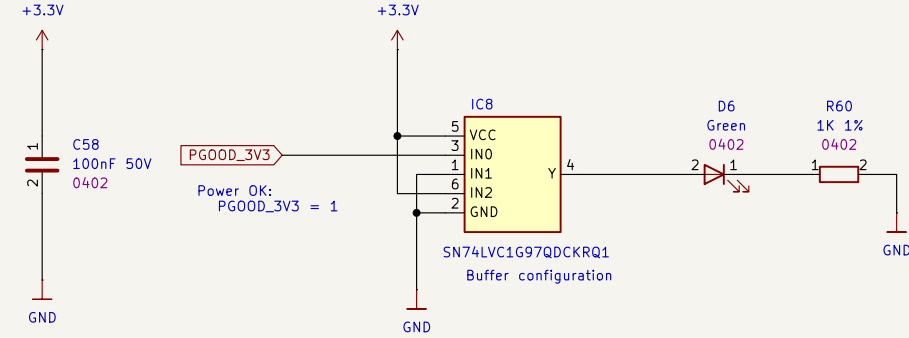
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	File Name: ethernet_interface.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 6 of 9

[7] USB Interface and Power

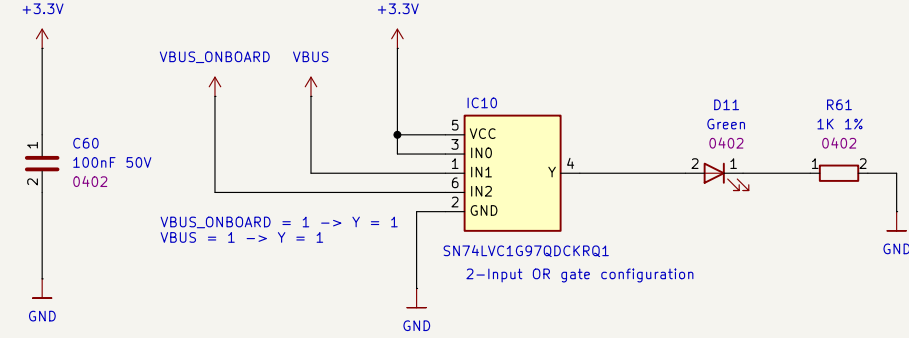
+5V to 3.3V PMIC



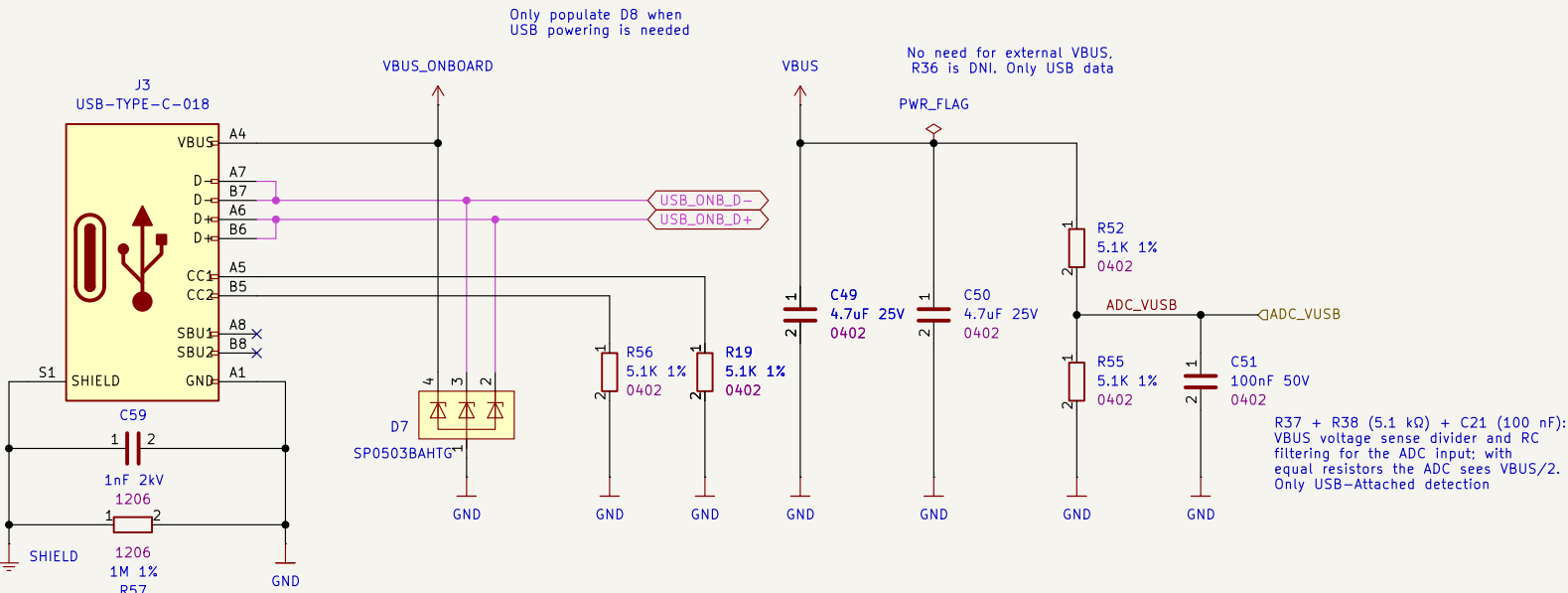
PGOOD LOGIC



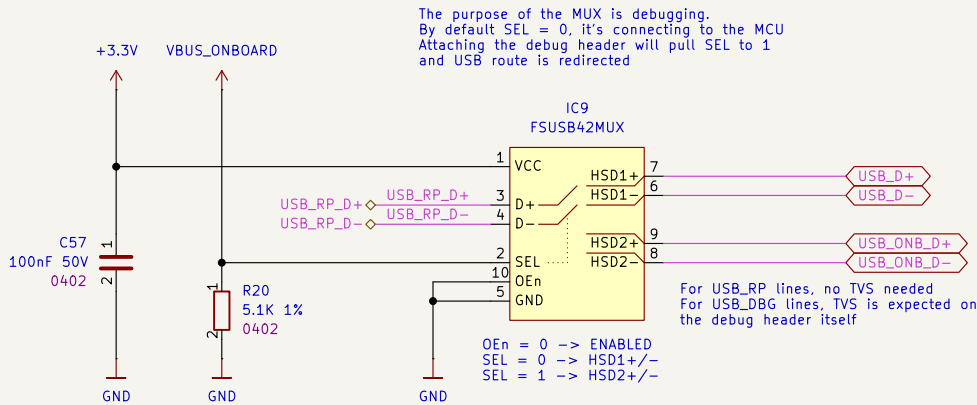
VBUS INDICATION LOGIC



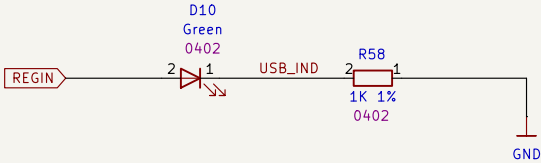
ONBOARD USB CONNECTOR



USB DATA-PATH SELECTOR

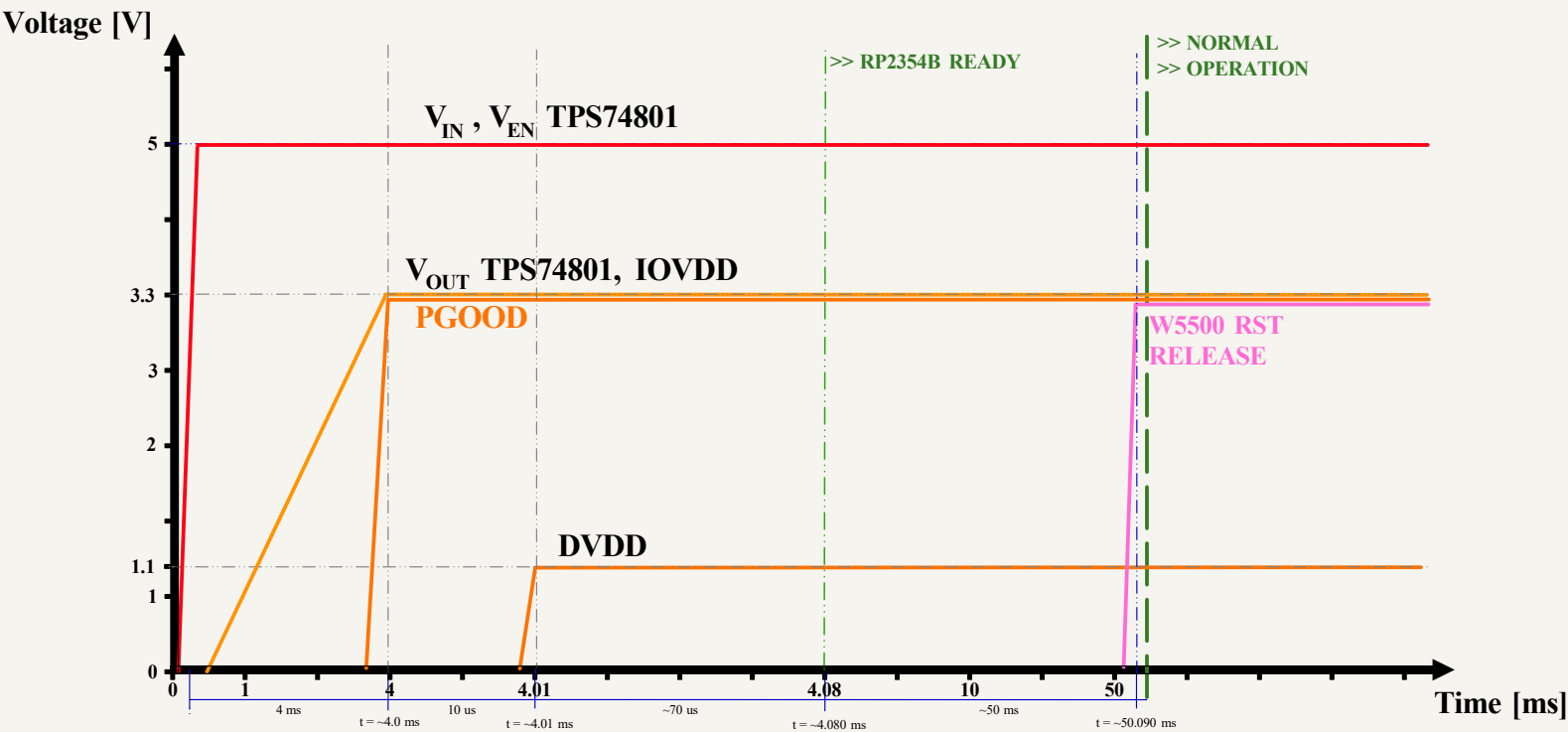


+5V INDICATOR LED



 Date: 2026-01-31 Sheet Title: USB Interface and Power	Board Name: BladeCore		Project Name: M54	
	File Name: usb_interface.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 7 of 9

[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms
7	RP2354B ready	0.080 ms	4.090 ms
8	W5500 ready	50.000 ms	54.090 ms



Date: 2026-01-31

Sheet Title:
Power - Sequencing

Board Name:
BladeCore

File Name:
Power - Sequencing.kicad_sch

Company:
DvidMakesThings

Designer:
David Sipos
Reviewer:

Project Name:
M54

Revision:
1.0.0

Size:
A3

Sheet:
8 of 9

[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

 Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant:
	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3 Sheet: 9 of 9